

Question ID: c22b5f25

| Assessment | Test | Domain  | Skill            | Difficulty                                          |
|------------|------|---------|------------------|-----------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Medium <div><div></div><div></div><div></div></div> |

Question

In the xy-plane, the points  $(-2,3)$  and  $(4,-5)$  lie on the graph of which of the following linear functions?

Answer

A.  $f(x) = x + 5$

B.  $f(x) = \frac{1}{2}x + 4$

C.  $f(x) = -\frac{4}{3}x + \frac{1}{3}$

D.  $f(x) = -\frac{3}{2}x + 1$

Correct Answer: C

Rationale

Choice C is correct. A linear function can be written in the form  $f(x) = mx + b$ , where m is the slope and b is the y-coordinate of the y-intercept of the line. The slope of the graph can be found using the formula  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . Substituting the values of the given points into this formula yields  $m = \frac{-5 - 3}{4 - (-2)}$  or  $m = \frac{-8}{6}$ , which simplifies to  $m = -\frac{4}{3}$ . Only choice C shows an equation with this slope.

Choices A, B, and D are incorrect and may result from computation errors or misinterpreting the given information.